

(1001CJA102120112)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
17-07-2021
JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A,B,C,D	B,D	D	A	A,B	B,C	B,C,D	B,C	A,B,C,D
	Q.	11	12	13	14						
	A.	B	A	A,C	B,D						
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	8.00	0.20 to 0.22	1.27	2.50	3.00	3.20	3.00	5.00	36.20 to 36.30	2.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,C,D	A,B,C	B,D	B,D	A,B,C	A,B,D	B	C	C	A
	Q.	11	12	13	14						
	A.	B	A,B,D	A	A						
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3.00	5.00	5.80	3.00	1.50	316.00	3266.00	5.00	0.50	8.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	B,D	A,B,C	A,B,D	A,B,D	A,B,C	A,B,D	A,B,C	B,C	A,B,C
	Q.	11	12	13	14						
	A.	A	A,B,D	C	D						
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	4.00	1.33 or 1.34	3.66 or 3.67	0.00	2630.00	0.00	0.50	4.00	4.20	72.00

(1001CJA102120113)

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SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	A,C	D	A,D	A	A	B	A	B
	Q.	11	12	13	14						
	A.	B	B	B	A						
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	20.00	1.57	9.00	4.50	0.25	0.83	4.00	6.32 to 6.33	7.00	8.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C,D	B	A,D	B,C,D	A,B,C,D	A,B,C	B	A,D	D	C
	Q.	11	12	13	14						
	A.	C	C	B	A						
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	12.50	12.00	10.00	33.33	100.00	3.00	20.00	4.00	6.00	7.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A,B,C	B,C	A,B,C,D	A,B,C,D	A,B,C,D	A,C,D	A,C	B	D
	Q.	11	12	13	14						
	A.	C	D	D	D						
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	7.00	1.00	1.50	8.00	2.00	2.50	1.50	0.60	0.50	0.15



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PAPER-1

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (C)

Sol. $180^\circ \quad 90$
 $X \longrightarrow 2 Y$

so energy released

$$Q = 2 \times [90 \times 6] - [180 \times 4] = 360 \text{ MeV}$$

2. Ans. (A, B, C, D)

Sol. (A) Light rays should come out parallel as object as focus and it should also go out as parallel beam as slab does not converge or diverge the rays.

(B) Same as (A)

(C) Second lens is a diverging lens.

(D) Parallel beam through diverging lens should meet at focal plane virtually.

3. Ans. (B, D)

Sol. For underflected motion,

$$q\vec{E} + q(\vec{v} \times \vec{B}) = 0$$

So,

$$q(-E\hat{k}) + q(\vec{v} \times B\hat{j}) = 0$$

(B) $q > 0$

$$= q \left[-E\hat{k} + \frac{E}{B} \hat{i} \times B\hat{j} \right] = q \left[-E\hat{k} + E\hat{k} \right] = 0$$

It is clear it does not depend on positive or negative nature of charge.

4. Ans. (D)

Sol. Centripetal force = Gravitational force :

$$\frac{GMm}{r^2} = m \frac{4\pi^2}{T^2} r \Rightarrow r = \left(\frac{GMT^2}{4\pi^2} \right)^{1/3}$$

For the moon, we have

$$r_M = \left(\frac{GMT_M^2}{4\pi^2} \right)^{1/3}$$

For the satellite, we have

$$r_S = \left(\frac{GMT_S^2}{4\pi^2} \right)^{1/3}$$

Therefore, we have

$$\frac{r_M}{r_S} = \left(\frac{GMT_M^2}{4\pi^2} \right)^{1/3} / \left(\frac{GMT_S^2}{4\pi^2} \right)^{1/3} = \left(\frac{T_M}{T_S} \right)^{2/3}$$

5. Ans. (A)

Sol. The efficiency of cyclic process is

$$\epsilon = \frac{\text{Work done in complete process}}{\text{heat absorbed}}$$

For isothermal process

$$\Delta Q = \Delta U + \Delta W$$

$$Q = \Delta + W$$

$$Q = W$$

For isochoric process work done is

$$= -\frac{P_0}{2} (2V_0 - V_0) = -\frac{P_0 V_0}{2}$$

$$= -\frac{1}{2} (RT_0) = -\frac{RT_0}{2}$$

(using $P_0 V_0 = 1 \times RT_0$)

for iso basic process work done is zero.

and heat absorbed is

$$Q = nC_v \Delta T = 1 \times \frac{3R}{2} \times \left(\frac{T_0}{2} \right) = \frac{3RT_0}{4}$$

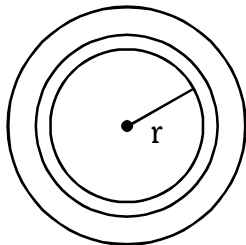
$$\epsilon = \frac{Q + \left(-\frac{RT_0}{2} \right)}{Q + \frac{3RT_0}{4}} = \frac{4Q - 2RT_0}{4Q + 3RT_0}$$

6. Ans. (A, B)

Sol. For radial flow conductivity should be spherical symmetrical as given.

In steady state

$$i_H = - \left(\frac{K_0}{r^2} \right) (4\pi r^2) \left(\frac{dT}{dr} \right)$$



$$i_H \int_0^R dr = -4\pi k_0 \int_{T_1}^{T_2} dt$$

$$i_H = \frac{4\pi K_0 (T_1 - T_2)}{R}$$

7. Ans. (B, C)

Sol. (A) $M = \int_0^L \lambda dx = \lambda_0 \int_0^L \left(1 + \frac{x^2}{L^2} \right) dx$

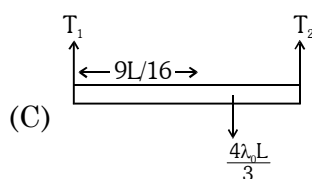
$$= \lambda_0 \left[L + \frac{1}{L^2} \times \frac{L^3}{3} \right] = \frac{4\lambda_0 L}{3}$$

(B) $X_{CM} = \frac{\int dm x}{M}$

$$= \frac{\int_0^L \lambda x dx}{M} = \frac{\lambda_0 \int_0^L \left(x + \frac{x^3}{L^2} \right) dx}{M}$$

$$= \frac{\lambda_0 \left[\frac{L^2}{2} + \frac{L^2}{4} \right]}{\frac{4\lambda_0 L}{3}} = \frac{3\lambda_0 L^2 \times 3}{4 \times 4\lambda_0 L}$$

$$X = \frac{9L}{16}$$



$$T_1 + T_2 = \frac{4\lambda_0 L g}{3}$$

$$T_1 \times \frac{gL}{16} = T_2 \times \frac{7L}{16}$$

$$\text{So, } T_1 = \frac{7\lambda_0 L g}{12} \quad T_2 = \frac{9\lambda_0 L g}{12}$$

8. Ans. (B,C,D)

Sol. For pure rolling

$$mg \sin \theta - f_s = ma$$

$$f_s r = \frac{mr^2}{2} \alpha$$

$$a = \alpha r$$

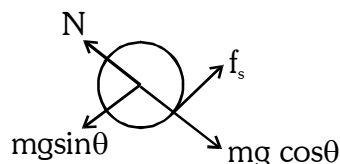
solve these question then

$$a = \frac{2g \sin \theta}{3}$$

$$f_s = \frac{mg \sin \theta}{3}$$

for no slipping

$$\frac{mg \sin \theta}{3} \leq \mu mg \cos \theta$$



(A) if $\mu < \frac{\tan \theta}{3}$, slipping will be there

$$\text{so, } a = g \sin \theta - \frac{f_k}{m} < g \sin \theta$$

$$(C) mgh = \frac{1}{2} mv^2 + \frac{1}{2} \frac{mv^2}{2} \left(\frac{v}{r} \right)^2$$

$$v = \sqrt{\frac{4gh}{3}}$$

9. Ans. (B, C)

Sol. Field at origin is only due to $-q$ and potential is more due to positive charges.

10. Ans. (A, B, C, D)

Sol. (A) At origin electric field is due to $-q$ charges.

(B) Just right of $-q$ charge the field is mainly due to $-q$ charge

(C,D) Far away from system, the system is like a $+q$ charge ($+2q - q = +q$) so field will be away from charge.

11. Ans. (B)

Sol. By continuity argument the points of zero electric field must be 2.

12. Ans. (A)

Sol. After removing middle spring both carts acts two spring block systems.

$$\frac{T_A}{T_B} = \frac{2\pi\sqrt{\frac{m}{k}}}{2\pi\sqrt{\frac{m}{k}}} = 1$$

13. Ans. (A,C)

Sol. If both carts oscillates in same phase the middle spring length remains fixed and effectively both blocks perform SHM with

one spring so angular frequency $\omega = \sqrt{\frac{k}{m}}$.

14. Ans. (B, D)

Sol. For the out of phase situation, the separation between the objects is not constant, but there exists a point on the connecting spring that does not move. As such, either object can be thought of moving under the influence of two springs in parallel, one with length L , the other with length $L/2$. That is equivalent to a spring with constant k in parallel with a spring of constant $2k$. The net spring constant is then $k + 2k = 3k$.

As such, the frequency is given by

$$\omega = \sqrt{3k/m}$$

SECTION-II

1. Ans. 8.00

Sol. Resulting wave is

$$y = 3\cos(kx - \omega t) + 5\cos(kx + \omega t)$$

The instantaneous power for a wave on string

$$P = -T \left(\frac{\partial y}{\partial t} \right) \left(\frac{\partial y}{\partial x} \right)$$

$$P = -\mu v^2 (+3\omega \sin(kx - \omega t) - 5\omega \sin(kx + \omega t))x$$

$$(-3k \sin(kx - \omega t) - 5k \sin(kx + \omega t))$$

$$P = \mu v^2 \omega k [9 \sin^2(kx - \omega t) - 25 \sin^2(kx + \omega t)]$$

$$\langle P \rangle = \mu v^2 \omega k [9 \langle \sin^2(kx - \omega t) \rangle - 25 \langle \sin^2(kx + \omega t) \rangle]$$

$$= \mu v^2 \omega k \left[\frac{9}{2} - \frac{25}{2} \right] = -8\rho v^2 \omega k$$

$$= 8\rho S \left(\frac{\omega}{k} \right)^2 \omega k = -\frac{8\rho S \omega^3}{k}$$

2. Ans. 0.20 to 0.22

Sol. Intensity at point 0.15 mm above central maxima in original system.

for that point

$$\Delta x = \frac{dy}{D} = \frac{2 \times 10^{-3} \times 0.15 \times 10^{-3}}{2}$$

$$= 1.5 \times 10^{-7} \text{ m}$$

phase difference

$$\Delta Q = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{6000 \times 10^{-10}} \times \frac{3}{2} \times 10^{-7} = \frac{\pi}{2}$$

Initially intensity is same so intensity

$$I_R = 2I_0 (1 + \cos \Delta \theta) = 2I_0$$

Now if a sheet is placed the Δx of that point

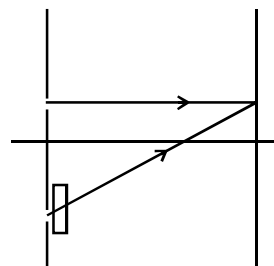
$$\Delta x = \frac{dy}{D} + (\mu - 1)t$$

$$= 1.5 \times 10^{-7} + \left(\frac{3}{2} - 1 \right) 8000 \times 10^{-10}$$

$$= 1.5 \times 10^{-7} + 4 \times 10^{-7} = 5.5 \times 10^{-7} \text{ m}$$

$$\text{phase diff. } \Delta Q = \frac{2\pi}{6000 \times 10^{-10}} \times \frac{11}{2} \times 10^{-7}$$

$$= \frac{11\pi}{6}$$



no sheet allow only η fraction so to get same intensity

$$I + \eta I + 2\sqrt{\eta} I \cos\left(\frac{11\pi}{6}\right) = 2I_0$$

$$\eta^2 - 5\eta + 1 = 0$$

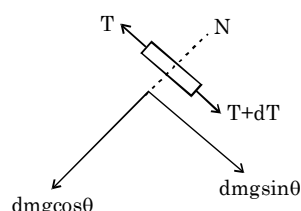
$$\eta = \frac{5 - \sqrt{21}}{2} = 0.21$$

3. Ans. 1.27

Sol. For any small element force analysis just after release.

$$dm g \sin \theta + dT = dma$$

mass of element can be written as



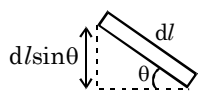
$$dm = \lambda dl$$

$$\lambda g dl \sin \theta + dT = dma$$

$$\lambda g dy + dT = dma$$

$$\lambda g \int_0^{4R} dy + \int_0^0 dT = a \int_0^M dm$$

$$\lambda g(4R) = Ma$$



$$\lambda g(4R) = \lambda(2\pi R)a$$

$$\therefore x = \frac{2}{\pi}$$

$$2 \times \frac{2}{\pi} = 1.27$$

4. Ans. 2.50

Sol. Initially, the velocity of the cube and rectangle block are v and 0 respectively.

The kinetic friction is $f_k = \mu mg$ and the acceleration on the cube and the block are

$$a_m = -\mu g \text{ and } a_M = \frac{\mu mg}{M}$$

$$v_m = -\mu gt$$

$$v_M = \frac{\mu mgt}{M}$$

They will accelerate until they have the same velocity when the cube is at the right end, i.e.,

$$v - \mu gt = \frac{\mu mgt}{M} \Rightarrow \left(\frac{m}{M} + 1\right) \mu gt = v$$

$$\Rightarrow t = \frac{v}{\mu g} \left(\frac{1}{1 + \frac{m}{M}} \right)$$

At this time the position of the cube and the right end of the block are:

$$x_m = vt + \frac{1}{2} a_m t^2 = \frac{v^2}{\mu g} \left(\frac{1}{1 + \frac{m}{M}} \right) - \frac{1}{2} \frac{v^2}{\mu g} \left(\frac{1}{1 + \frac{m}{M}} \right)^2$$

$$= \frac{v^2}{\mu g} \left(\frac{1}{1 + \frac{m}{M}} \right) \left(1 - \frac{1}{2} \left(\frac{1}{1 + \frac{m}{M}} \right) \right)$$

$$x_m = \frac{v^2}{\mu g} \left(\frac{\frac{1}{2} + \frac{m}{M}}{\left(1 + \frac{m}{M} \right)^2} \right)$$

$$x_m = L + \frac{1}{2} a_M t^2 = L + \frac{1}{2} \frac{m}{M} \frac{v^2}{\mu g} \left(\frac{1}{1 + \frac{m}{M}} \right)^2$$

If they coincide at time t ,

$$x_m = x_M$$

$$\Rightarrow \frac{v^2}{\mu g} \frac{1}{\left(1 + \frac{m}{M} \right)^2} \left(\frac{1}{2} + \frac{m}{M} - \frac{1}{2} \frac{m}{M} \right) = L$$

$$\Rightarrow \frac{v^2}{2\mu g} = L \left(1 + \frac{m}{M} \right) \Rightarrow v = \sqrt{2\mu g L \left(1 + \frac{m}{M} \right)}$$

The displacement of the block is

$$d = \frac{1}{2} a_M t^2 = \frac{1}{2} \frac{m}{M} \frac{v^2}{\mu g} \left(\frac{1}{1 + \frac{m}{M}} \right)^2$$

$$= \frac{m}{2M\mu g} \left(\frac{1}{1 + \frac{m}{M}} \right)^2 \times 2\mu g L \left(1 + \frac{m}{M} \right)$$

$$d = \frac{L \frac{m}{M}}{1 + \frac{m}{M}} = \left(\frac{m}{m+M} \right) L$$

$$= \frac{2 \times 10}{2+6} = \frac{20}{8} = 2.50 \text{m}$$

5. **Ans. 3.00**

Sol. After collision both atoms are moving together so collision between them must be completely inelastic and maximum loss in this collision is half of initial KE of atom. so to excite both after collision minimum must be $(10.2 + 10.2) = 20.4 \text{eV}$

$$\frac{K}{2} \geq 20.4$$

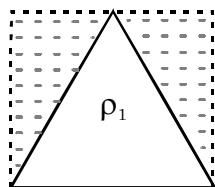
$$K \geq 2 \times 20.4$$

$$K \geq 40.8$$

$$\text{So } x = 3.00$$

6. **Ans. 3.20**

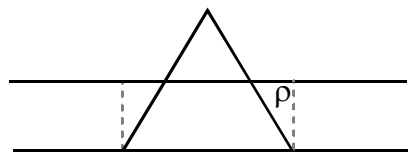
Sol. For equilibrium force on cone due to ρ_1 and ρ_2 must be equal for force of ρ_1



upward force due to ρ_1 is equal to weight of liquid in shaded region.

$$F_{\text{up}} = \frac{2}{3} \pi H^2 (2H) \rho_1 g$$

Downward force of ρ_2 is equal to weight of ρ_2 above cone



$$F_{\text{down}} = \rho_2 v g \text{ where } v \text{ is volume of } \rho_2$$

Volume ρ_2 is

$$v = \pi H^2 \times H - v_{\text{cone}}$$

Here V_{cone} is volume of cone up to height H from base.

$$v_{\text{cone}} = v_{\text{total}} - v'_{\text{cone}}$$

$$v_{\text{total}} = \text{total volume of cone} = \frac{1}{3} \pi H^2 (2H)$$

v'_{cone} = volume of cone above

$$\rho_2 = \frac{1}{3} \pi \left(\frac{H}{2} \right)^2 H = \frac{1}{3} \pi \left(\frac{H}{2} \right)^2 H$$

$$v_{\text{cone}} = \frac{2}{3} \pi H^3 - \frac{\pi H^3}{12} = \frac{7}{12} \pi H^3$$

$$\text{Now } v = \pi H^3 - \frac{7}{12} \pi H^3 = \frac{5}{12} \pi H^3$$

$$F_{\text{down}} = \rho_2 \left(\frac{5}{12} \pi H^3 \right) g$$

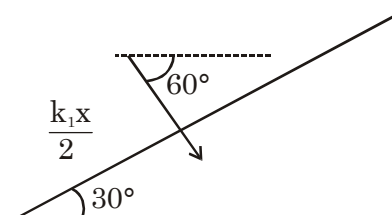
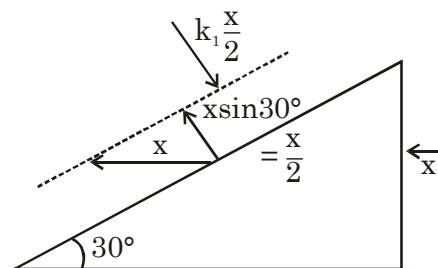
for equilibrium

$$F_{\text{up}} = F_{\text{down}}$$

$$\frac{\rho_2}{\rho_1} = \frac{16}{5}$$

7. **Ans. 3.00**

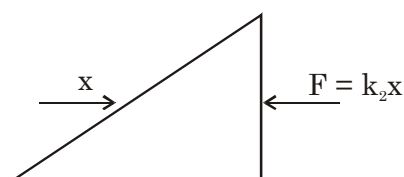
Sol. (A) On displacing the wedge by distance x towards left



$$\therefore F_{\text{net}} = \left(\frac{k_1 \cos 60^\circ}{2} \right) x = \left(\frac{k_1}{4} \right) x$$

$$\text{Half time period} = \pi \sqrt{\frac{m}{\frac{k_1}{4}}} = \pi \sqrt{\frac{4m}{k_1}}$$

(B) On displacing towards right



$$F_{\text{net}} = k_2 x \Rightarrow \text{Half time period} = \pi \sqrt{\frac{m}{k_2}}$$

(C) So, total time period

$$\begin{aligned} & \pi \sqrt{\frac{4m}{k_1}} + \pi \sqrt{\frac{m}{k_2}} \\ &= \pi \sqrt{\frac{4 \times 4}{16\pi^2}} + \pi \sqrt{\frac{4}{\pi^2}} \\ &= 1 + 2 = 3 \text{ seconds} \end{aligned}$$

8. Ans. 5.00

9. Ans. 36.20 to 36.30

$$\text{Sol. } -13.6Z^2 \left[\frac{1}{3^2} - \frac{1}{2^2} \right] = 47.5$$

$$13.6Z^2 \left[\frac{9-4}{36} \right] = 47.5$$

$$13.6Z^2 = \frac{47.5}{5} \times 36 = 9.5 \times 36 \text{ eV}$$

Energy required to ionize = $13.6 Z^2$

$$\Rightarrow \frac{hc}{\lambda} = 13.6Z^2$$

$$\therefore h = \frac{hc}{13.6Z^2} = \frac{hc}{9.5 \times 36}$$

$$= \frac{12400}{9.5 \times 36} \text{ \AA}$$

$$\lambda = 36.25 \text{ \AA}$$

10. Ans. 2.00

Sol. At time 't'

$$-L \frac{di}{dt} - iR = 0$$

$$-L \frac{di}{dt} = iR = V = 10i^2$$

$$-\int_{2.5}^{1.25} \frac{di}{i^2} = \frac{10}{L} \int_0^t dt$$

$$\Rightarrow \left(\frac{1}{L} \right)_{2.5}^{1.25} = \frac{10}{50 \times 10^{-3}} [t - 0]$$

$$\Rightarrow \frac{4}{5} - \frac{2}{5} = \frac{1000}{5} \times t$$

$$\therefore t = \frac{2}{1000} = 2 \text{ ms}$$

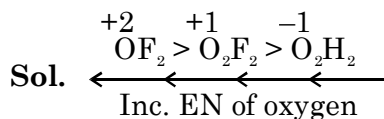
PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (B,C,D)

2. Ans. (A,B,C)



Fluorine has greater electron affinity than Be and N.

3. Ans. (B,D)

Sol. 2M MgCl_2

$$\text{molality} = \frac{\text{moles of } \text{MgCl}_2}{\text{mass of solvent in kg}}$$

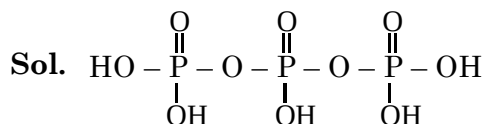
$$= \frac{2}{\frac{900}{1000}} = 4.44 \text{ m}$$

$$\text{Mole fraction of } \text{MgCl}_2 = \frac{n_{\text{MgCl}_2}}{n_{\text{MgCl}_2} + n_{\text{H}_2\text{O}}}$$

$$\begin{aligned} &= \frac{2}{\frac{900}{18} + 2} = \frac{2}{52} \\ &= 0.038 \end{aligned}$$

4. Ans. (B,D)

5. Ans. (A,B,C)



Two P - O - P linkages are present.

$\text{H}_2\text{S}_{n+2}\text{O}_6$ (n+1) S - S linkages.

6. Ans. (A,B,D)

Sol. $2x + 3y + 4z \rightarrow 5w$

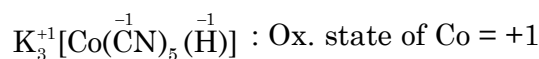
$$\begin{array}{ccccccc} & 2x & + & 3y & + & 4z & \rightarrow 5w \\ t=0 & 1 \text{ mole} & & 3 \text{ mole} & & 4 & \end{array}$$

% yield = $\frac{\text{observed value}}{\text{theoretical value}}$

$$= \frac{1.25}{2.50} \times 100 = 5$$

7. Ans. (B)

Sol. $\text{Li}[\text{CuMe}_2]$: Ox.state of Cu = +1



$\text{In}[\text{Co}(\text{en})_3]\text{Cl}_3$, en is ethane-1,2-diamine not diammine.

8. Ans. (C)

Sol. Serpeck's process is used for concentration of white bauxite. Leaching of ore with NaOH produces NaAlO_2 in filtrate not as ppt.

9. Ans. (C)

10. Ans. (A)

Sol. $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$

$$t = 0 \quad \text{CM}$$

$$t \quad \text{C}-\text{C}\alpha \quad \text{C}\alpha \quad \text{C}\alpha$$

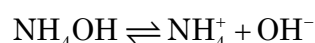
$$K_a = \frac{(\text{C}\alpha)(\text{C}\alpha)}{(\text{C}-\text{C}\alpha)} = \frac{\text{C}\alpha^2}{1-\alpha}$$

$$\text{if } \alpha \ll 1$$

$$K_a = \text{C}\alpha^2$$

$$\alpha = \sqrt{\frac{K_a}{\text{C}}}$$

$$\Rightarrow [\text{H}^+] = \text{C}\alpha = \sqrt{K_a \times \text{C}} = \sqrt{10^{-5} \times 10^{-1}} = 10^{-3}$$

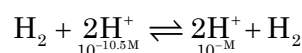


$$K_b = \frac{\text{C}\alpha^2}{1-\alpha}$$

$$\text{if } \alpha \ll 1$$

$$[\text{OH}^-] = \text{C}\alpha = \sqrt{K_b \times \text{C}} = \sqrt{10^{-5} \times 10^{-2}}$$

$$= \sqrt{10^{-7}} = 10^{-3.5} \text{ M}$$



$$E = E^\circ - \frac{0.059}{2} \log \frac{(10^{-3})^2}{(10^{-10.5})^2}$$

$$E = \frac{-0.059}{2} \log \frac{10^{-6}}{10^{-21}}$$

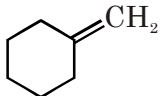
$$= \frac{-0.059}{2} \times 15 = -0.44 \text{ V}$$

11. Ans. (B)

12. Ans. (A,B,D)

13. Ans. (A)

14. Ans. (A)

Sol. Compound A is  and
(4 α -H)

Compound B is $\text{CH}_2=\text{CH}_2$ (no α -H)

SECTION-II

1. Ans. 3.00

Sol. $\text{A} \rightarrow \text{B}$

$$K_1 = A_1 \times e^{\frac{-E_{a1}}{RT}}$$

$$\text{x} \rightarrow \text{y}$$

$$K_2 = A_2 \times e^{\frac{-E_{a2}}{RT}}$$

$$\text{if } k_1 = k_2$$

$$10^9 e^{\frac{-1200}{RT}} = 10^{10} e^{\frac{-1800}{RT}}$$

$$\frac{-1200}{RT} = \frac{-1800}{RT} + \ln 10$$

$$T = \frac{300}{2.303} \text{ K}$$

2. Ans. (5.00)

Sol. $\Delta E = \Delta q - w$

$$\Delta E = -w = -P [V_2 - V_1]$$

$$\Delta E = -5 [2\text{L}]$$

$$\Delta E = -10 \text{ L} \cdot \text{atm}$$

3. Ans. 5.80

Sol. $x = 5$

$$y = 2$$

$$z = 5$$

4. Ans. 3.00

Sol. $y = 78$ (benzene)

$$x = 26 \text{ (acetylene)}$$

5. Ans. 1.50

Sol. $2\text{Na} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{H}_2$

$$X = 0.50$$



$$Y = 1$$

$$X + Y = 1.50$$

6. Ans. 316.00

Sol. $3\text{K}_2\text{MnO}_4 + 2\text{H}_2\text{SO}_4 \rightarrow 2\text{KMnO}_4 + \text{MnO}_2 + 2\text{K}_2\text{SO}_4 + 2\text{H}_2\text{O}$

$$\text{mass of Mn compounds} = 2 \times \frac{158}{\text{KMnO}_4}$$

$$= 316.00 \text{ gram}$$

7. Ans. (3266.00)

8. Ans. (5.00)

Sol. $x = 2$

$$y = 3$$

(i) Aromatic aldehyde and Ketones do not react with Fehling solution.

(ii) CH_3CHO & HCOOH Both compounds give positive tollen's test.

9. Ans. 0.50

10. Ans. 8.00

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,C)**

Sol. S_1 and S_2 touch each other externally
Equation of direct common tangents

$$y = m \left(x - \frac{8}{3} \right) \text{ as point P is point of intersection of direct common tangents } \left(\frac{8}{3}, 0 \right)$$

$$\text{condition of tangency } \frac{\left(m - \frac{8m}{3} \right)}{\sqrt{1+m^2}} = 1$$

$$\Rightarrow m^2 = \frac{9}{16}$$

2. **Ans. (B,D)**

Sol. (A) $\lim_{x \rightarrow 0^+} x^x = 1$

$$(B) -\frac{1}{2} \lim_{x \rightarrow 0^+} x^2 \ln x = -\frac{1}{2} \lim_{x \rightarrow 0^+} \frac{1/x}{-2/x^3} = 0$$

$$(C) \ell = \lim_{x \rightarrow 0^+} x^{\ln(x+1)}$$

$$\Rightarrow \ln \ell = \lim_{x \rightarrow 0^+} \ln(x+1) \ln x = 0$$

$$\Rightarrow \ell = 1$$

$$(D) \lim_{x \rightarrow 0^+} \frac{\left(\frac{5^x - 1}{x} \right) \left(\frac{2^x - 1}{x} \right) x}{1 + \frac{\tan x}{x}} = 0$$

3. **Ans. (A,B,C)**

$$\text{Sol. } S_n = \sum_{n=1}^n \frac{2n+1}{n^2(n+1)^2} = 1 - \frac{1}{(n+1)^2}$$

$$P_n = (n+1)^2$$

$$P_{40} = (41)^2 = 1681$$

$$P_{50} = (51)^2 = 2601$$

4. **Ans. (A,B,D)**

$$\text{Sol. (A) } \sqrt{x} + \sqrt{y} = 1$$

$$\Rightarrow x^2 - 2xy + y^2 - 2x - 2y + 1 = 0$$

$$(B) \frac{1}{x} + \frac{1}{y} = 1 \text{ hyperbola}$$

$$(C) \frac{1}{x} + \frac{1}{y} + \frac{1}{x+y} = 0$$

$$\Rightarrow x^2 + 3xy + y^2 = 0 \text{ pair of lines}$$

$$(D) x + y - \frac{1}{x+y} = 0 \text{ pair of lines}$$

5. **Ans. (A,B,D)**

Sol. Let $f(x_1) = f(x_2)$

$$\Rightarrow f(f(f(x_1))) = f(f(f(x_2))) \Rightarrow x_1 = x_2$$

So, f is one-one

as $f(x)$ is continuous & one-one then it will be either strictly increasing or strictly decreasing.

If $x_1 < x_2$ and f is decreasing

then $f(x_1) > f(x_2)$

$$\Rightarrow f(f(x_1)) < f(f(x_2))$$

$$\Rightarrow f(f(f(x_1))) > f(f(f(x_2)))$$

$$\Rightarrow x_1 > x_2 \text{ (which contradicts)}$$

So, f is increasing

$$\text{Now, say } f(x) \geq x \Rightarrow f(f(x)) \geq f(x) \geq x$$

$$\Rightarrow f(f(f(x))) \geq f(x)$$

$$\Rightarrow x \geq f(x)$$

So, $f(x) = x$

$$\text{Similarly if } f(x) \leq x \Rightarrow f(x) \geq x$$

$$\Rightarrow f(x) = x$$

6. **Ans. (A,B,C)**

$$\text{Sol. Equation of mirror } \Rightarrow y - 4 = \frac{4}{3}(x - 1)$$

$$\Rightarrow 4x - 3y + 8 = 0$$

$$\text{Point D } \left(\frac{1}{3}, \frac{1}{3} \right)$$

Equation of angle bisector of

$$\angle A \Rightarrow x - 7y + 2 = 0$$

Point E (-2, 0)

$$\Delta CDE = \frac{1}{2} \times 5 \times \frac{5}{3} = \frac{25}{6}$$

$$\text{length BC} = 5\sqrt{5}, \text{ Equation of}$$

$$\text{BC } 11x - 2y - 3 = 0$$

$$\Delta ABC = \frac{1}{2} \cdot 5\sqrt{5} \cdot 2\sqrt{5} = 25$$

$$\Delta BCE = \frac{1}{2} \cdot 5\sqrt{5} \cdot \sqrt{5} = \frac{25}{2}$$

7. Ans. (A,B,D)

Sol. $[\sin^{-1}x] \in \{-2, -1, 0, 1\}$

$[\cos^{-1}x] \in \{0, 1, 2, 3\}$

case 1 $1 + [\sin^{-1}x] = 1$ and $[\cos^{-1}x] = 0$

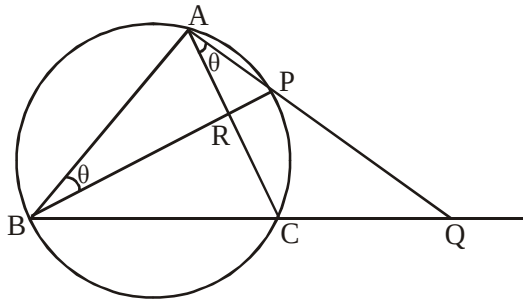
$0 \leq x < \sin 1$ and $\cos 1 < x \leq 1$

case 2 $1 + [\sin^{-1}x] = 2$ and $[\cos^{-1}x] = 0$ or 1

$\sin 1 \leq x \leq 1$ and $\cos 2 < x \leq 1$

$\Rightarrow \sin 1 \leq x \leq 1$

8. Ans. (A,B,C)



Sol. $\frac{AB}{\sin B} = \frac{b}{\sin B/2}$

$\Rightarrow AQ = 2b \cos B/2$

and PQ. $AQ = QC \cdot QB$

$\Rightarrow PQ = \frac{QC}{QA} \cdot QB$

In $\triangle AQC$

$$\frac{QC}{\sin B/2} = \frac{QA}{\sin B} \Rightarrow \frac{QC}{QA} = \frac{1}{2 \cos B/2}$$

In $\triangle ABQ$ $BQ = \frac{c \sin(A + B/2)}{\sin B/2}$

$$PQ = \frac{c \sin(A + B/2)}{\sin B}$$

Paragraph for question no. 9 to 11

9. Ans. (B,C)

Sol. Option (A) $\Rightarrow [3, 4, 5] \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix} = [21 \ 20 \ 32]$

10. Ans. (A,B,C)

Sol. $|A| = 1 = |B| = |C|$

So $|AB| = 1 = |BC| = |CA| = |ABC|$

11. Ans. (A)

Sol. $\text{Tr}(A+B^T+3C) = 5 + 3 + 9 = 17$

Paragraph for question no. 12 to 14

12. Ans. (A,B,D)

13. Ans. (C)

14. Ans. (D)

Sol. $P(E_i) = K_i (i+1)$

$$\sum P(E_i) = 1 \Rightarrow K = \frac{3}{n(n+1)(n+2)}$$

$$P(E) = \sum_{i=1}^n P(E_i) P(E/E_i)$$

$$= \sum K_i (i+1) \cdot \frac{i}{n} = \frac{(3n+1)(n+2)}{4n(n+2)}$$

When $n \rightarrow \infty$ $P(E) = 3/4$

$$P\left(\frac{E_1}{E}\right) = \frac{K \cdot 2 \cdot \frac{1}{n}}{\frac{(3n+1)(n+2)}{4n(n+2)}} = \frac{24}{n(n+1)(n+2)(3n+1)}$$

SECTION-II

1. Ans. 4.00

Sol. $\frac{x}{x_1} + \frac{y}{y_1} = 2$, $MQ = \frac{C^2}{2x_1}$

$$MV = 2y_1 = \frac{2c^2}{x_1}, \text{ hence } \lambda = 4$$

2. Ans. 0.33 or 0.34

Sol. $\frac{2f(x)f'(x)}{\sqrt{1-(f^2(x))^2}} \geq 2x$

$$\Rightarrow \int_{x_1}^{x_2} \frac{2f(x)f'(x)}{\sqrt{1-(f^2(x))^2}} dx \geq \int_{x_1}^{x_2} 2x dx$$

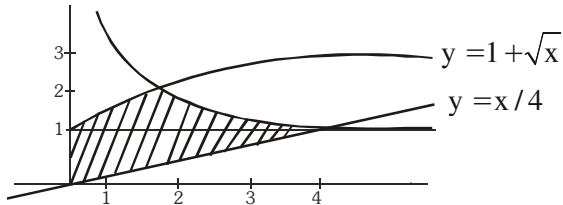
$$\Rightarrow (x_2^2 - x_1^2) \leq \sin^{-1}(f^2(x)) \Big|_{x_1}^{x_2}$$

$$\Rightarrow x_1^2 - x_2^2 \geq \frac{\pi}{3}$$

3. Ans. 3.66 or 3.67

$$\text{Sol. } A_1 = \int_0^1 \left(1 + \sqrt{x} - \frac{x}{4}\right) dx = \frac{37}{24}$$

$$\text{and } A_2 = \int_1^4 \left(\frac{2}{\sqrt{x}} - \frac{x}{4}\right) dx = \frac{17}{8}$$



4. Ans. (0.00)

$$\text{Sol. } \vec{c} = \vec{a} + \vec{b} \Rightarrow \vec{a} + \vec{b} - \vec{c} = 0$$

$$\Rightarrow [\vec{a} \vec{b} \vec{c}] = 0$$

5. Ans. 2630.00

$$\text{Sol. } (15\lambda + 1, 16\lambda - 1, n\lambda + 1) \text{ point}$$

$$\text{then } n\lambda + 1 = 0$$

$$6(15\lambda + 1)^2 + 5(16\lambda - 1)^2 = 1$$

$$\Rightarrow 6\left(1 - \frac{15}{n}\right)^2 + 5\left(-1 - \frac{16}{n}\right)^2 = 20$$

$$\Rightarrow 6(n - 15)^2 + 5(n + 16)^2 = 20n^2$$

$$\Rightarrow -9n^2 - 20n + 2630 = 0$$

6. Ans. 0.00

$$\text{Sol. } y = x - [x] = \{x\}$$

then

$$(K-2)y^2 + 2y + k^2 = 0, \text{ where } 0 < y < 1$$

x is not an integer

$$\text{for } 3 < x < 4 \Rightarrow [x] = 3$$

y has exactly one solution is $(0, 1)$

$$f(0), f(1) < 0 \Rightarrow K \in (-1, 0)$$

7. Ans. 0.50

$$\text{Sol. As } |f(x)| \leq 2$$

$$\text{We have } (f(x) - 2) \leq 0$$

$$\Rightarrow (f(x) - 2)(f(x) + 1)^2 \leq 0$$

$$\Rightarrow (f(x))^3 - 3f(x) - 2 \leq 0 \quad \forall x \in [0, 1]$$

$$\Rightarrow \int_0^1 (f(x))^3 dx \leq 3 \int_0^1 f(x) dx + 2 \int_0^1 dx$$

$$\Rightarrow \int_0^1 (f(x))^3 dx \leq 2$$

$$\Rightarrow \frac{1}{n} = 0.50$$

8. Ans. 4.00

$$\text{Sol. } z = (1 + ai)(1 + bi)(1 + ci)$$

$$|z|^2 = (1 + a^2)(1 + b^2)(1 + c^2)$$

$$= 25 \Rightarrow |z| = 5$$

$$\text{We have } -|z| \leq \text{Re}(z) \leq |z|$$

$$\Rightarrow -5 \leq 1 - ab - bc - ca \leq 5$$

$$\Rightarrow -4 \leq ab + bc + ca \leq 6$$

$$\lambda = -4, \text{ at } a = 2, b = -2, c = 0$$

9. Ans. 4.20

$$\text{Sol. } a_1 = 1, a_2 = 0, a_3 = 1,$$

$$a_n = a_{n-2} + a_{n-3}, n > 3$$

$$a_4 = 1, a_5 = 1, a_6 = 2, a_7 = 2$$

$$a_8 = 3, a_9 = 4, a_{10} = 5, a_{11} = 7$$

$$a_{12} = 9, a_{13} = 12, a_{14} = 16$$

10. Ans. 72.00

$$\text{Sol. As } y = 12, y = -12$$

meets $y = P(x)$ in 6 distinct points, least degree of $P(x)$ is 3

$$\Rightarrow P(x) - 12 = -2(x + 3)(x - 1)(x - 2)$$

$$\text{or } P(x) + 12 = 2(x + 3)(x - 1)(x - 2)$$

$$\Rightarrow P(4) = \pm 72$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

17-07-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-2

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (C)

Sol. Consider the frame moving with the final velocity of the plank (it is also the frame of mass center). In this frame initial kinetic energy of the block is completely consumed into the work done against the friction force and spring force.

$$V_{cm} = m v_0 / M + m,$$

Velocity of particle in center of mass frame

$$= \frac{M v_0}{M + m}$$

$$\frac{1}{2} m \left[\frac{m v_0}{M + m} \right]^2 = \text{energy stored in spring} + \text{work done against friction.}$$

$$v_0 = \mu g \sqrt{\frac{3m(m+M)}{kM}} = 0.4 \text{ m/s}$$

2. Ans. (B)

Sol. $h\nu = 17 \text{ eV}$

Now for metal,

$$h\nu = \phi + (KE)_{\max}$$

$$\Rightarrow 17 \text{ eV} = \phi + 2 \text{ eV} \Rightarrow \phi = 15 \text{ eV}$$

3. Ans. (A, C)

Sol. $\tau = NiAB = I\alpha$

$$N \cdot \frac{dq}{dt} AB = I \cdot \alpha$$

$$NAB \int_0^Q dq = I \int_0^\omega \alpha dt$$

$$NAB Q = I\omega$$

$$\omega = \frac{NABQ}{I}$$

$$\text{Kinetic energy} = \frac{1}{2} I \omega^2 = \frac{1}{2} C Q_0^2$$

$$\text{Maximum Deflection } (Q_0) = \omega \sqrt{\frac{I}{C}}$$

$$= \frac{NABQ}{\sqrt{IC}}$$

4. Ans. (D)

$$\text{Sol. } \lambda_A = \frac{0.693}{5} \text{ day}^{-1} \text{ and } \lambda_B = \frac{0.693}{30} \text{ day}^{-1}$$

$$\text{Given } \left(\frac{dN_{OA}}{dt} \right) = 64 \left(\frac{dN_{OB}}{dt} \right)$$

$$\therefore \lambda_A N_{OA} = 64 \lambda_B N_{OB}$$

$$\text{At time } t, \frac{dN_A}{dt} = \frac{dN_B}{dt}$$

$$\therefore \lambda_A N_A = \lambda_B N_B$$

Dividing equation (1) and (2)

$$\frac{N_{OA}}{N_A} = 64 \left(\frac{N_{OB}}{N_B} \right),$$

$$\text{But } N_A = N_{OA} e^{-\lambda_A t} \text{ and } N_B = N_{OB} e^{-\lambda_B t}$$

$$\therefore \text{equation (3) becomes } e^{\lambda_A t} = 64 e^{\lambda_B t}$$

$$\therefore e^{(\lambda_A - \lambda_B)t} = 64$$

$$\therefore (\lambda_A - \lambda_B)t = \ln(64) = 2.303 \log_{10} 64$$

$$\therefore t = \frac{2.303 \log_{10} 64}{\left(\frac{0.693}{5} - \frac{0.693}{30} \right)} = 36 \text{ days}$$

5. Ans. (A, D)

Sol. Point A and C are on the same line passing

$$\text{through origin} \Rightarrow \frac{P_A}{V_A} = \frac{P_C}{V_C} \dots (1)$$

$$\text{Also } T_A = 200 \text{ K} = \frac{P_A V_A}{nR} \text{ and Also } T_C = 1800$$

$$K = \frac{P_C V_C}{nR} \Rightarrow \frac{P_A V_A}{P_C V_C} = \frac{1}{9} \dots (2)$$

$$\text{From (1) and (2) } \frac{V_A}{V_C} = \frac{1}{3}$$

6. Ans. (A)

Sol. $v^2 = I \Rightarrow (v_0 - Ir)^2 = I$

7. Ans. (A)

Sol. We can choose an arbitrary center of rotation to solve this problem. In order to simplify our solution, let us choose the center of rotation so that the torque due to static friction is zero. That is accomplished by picking the point of contact with the ground as the center of rotation, because the lever arm of static friction is then 0. The magnitude of the torque due to force T around the contact point with the ground is ($\tau_1 = T \cdot 2a$). We want it to match the torque caused by the hole. To more experienced problem solvers it will be obvious that

$$\tau_2 = W \cdot \frac{2}{5}a,$$

where W is the weight of the removed material. Let us derive this for those less experienced. Let us denote the point of constant by C, the center of the cross section of the cylinder by O, and the center of the hole by M. By definition, the magnitude of torque τ_2 is

$$\tau_2 = W \cdot |CM| \cdot \sin\theta,$$

where $\theta = \angle(CO, CM)$. Since $|CM| \cdot \sin\theta = \frac{2a}{5}$, we see that indeed

$$\tau_2 = W \cdot \frac{2}{5}a,$$

Therefore,

$$\tau_1 = \tau_2 \Rightarrow 2aT = \frac{2aW}{5}$$

$$\Rightarrow T = \frac{W}{5} = 3\text{Nm}$$

so the closest answer is A.

8. Ans. (B)

Sol. $\frac{1}{V} - \frac{1.5}{\infty} = \frac{1-1.5}{-5}$

$$V = 10 \text{ cm}$$

9. Ans. (A)

10. Ans. (B)

11. Ans. (B)

Sol. (9 to 11)

$$I = \frac{V}{R} \text{ in saturation}$$

As $L_3 = L_4$ so currents in both of them is of

$$\text{equal magnitude and is } \frac{V}{2R}$$

$$\text{Current through } L_1 = \frac{IL_2}{L_1 + L_2}$$

$$\Rightarrow I_1 = \frac{VL^2}{R(L_1 + L_2)}$$

12. Ans. (B)

13. Ans. (B)

14. Ans. (A)

Sol. (12 to 14)

$$\lambda = \frac{V}{f} = \frac{330}{440} = \frac{3}{4}$$

$$\text{In second overtone } \ell = \frac{5\lambda}{4} = \frac{15}{16} \text{ m}$$

Let upper end be at $x = 0$

$$\therefore \Delta P = \Delta P_0 \sin \frac{2\pi x}{\lambda}$$

$$\text{At } x = \frac{L}{2}, \Delta P = \frac{\Delta P_0}{\sqrt{2}}$$

SECTION-II

1. Ans. 20.00

Sol. $P_0 - \rho g(4\text{mm}) + \rho g(3\text{mm}) + \frac{2s}{R} = P_0$

$$\Rightarrow \frac{2s}{R} = \rho g(1\text{mm})$$

$$\Rightarrow R = 200 \text{ mm}$$

2. Ans. 1.57

Sol. $Q_{\max} = \sigma_{\max} 4\pi R^2 = (\epsilon_0 E_0) 4\pi R^2 = 4\pi \epsilon_0 E_0 R^2$

$$F = \frac{Q^2}{32\pi \epsilon_0 R^2} \Rightarrow F = \frac{\pi \epsilon_0 E_0^2 R^2}{2}$$

3. Ans. 9.00

Sol. Since water resistance ($\vec{F} = -k\vec{v}_{pr}$) is proportional to the velocity of plank relative to the water, impulse of water

resistance equals to product of the proportionality constant k and displacement of the plank relative to the water. And this impulse stops the block-plank system at the other bank, therefore equals to the initial momentum of the block.

$$mu_{\min} = k \int v dt$$

$$\Rightarrow mu_{\min} = k(b-l)$$

$$u_{\min} = \frac{k(b-l)}{m} = 9.0 \text{ m/s}$$

4. **Ans. 4.50**

$$\text{Sol. } W_b = \frac{9}{2} C \varepsilon^2, \Delta U = \frac{9}{4} C \varepsilon^2, \text{ So } \frac{9}{4} C \varepsilon^2 = \frac{1}{2} LI^2,$$

$$I_{\max} = 3\varepsilon \sqrt{\frac{C}{2L}}$$

5. **Ans. 0.25**

$$\text{Sol. } z_{RL} = \sqrt{R^2 + \omega^2 L^2}$$

$$z_{RLC} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}$$

$$\cos \phi_{RL} = \frac{R}{\sqrt{R^2 + \omega^2 L^2}}$$

$$\cos \phi_{RLC} = \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}}$$

According to question

$$(\omega L)^2 = \left(\omega L - \frac{1}{C\omega} \right)^2 \Rightarrow C = \frac{1}{2\omega^2 L}$$

6. **Ans. 0.83**

$$\text{Sol. } \frac{\lambda_0 g L^3}{3YA}$$

7. **Ans. 4.00**

$$\text{Sol. } \frac{hc}{\lambda} = 13.6(3)^2 \left(\frac{1}{3^2} - \frac{1}{4^2} \right)$$

$$\frac{hc}{\lambda} = 13.6 \times \frac{7}{16} = KE_{\max} + \phi$$

$$V_{\max} = \frac{qBR}{m}$$

$$KE_{\max} = \frac{1}{2} mv_{\max}^2$$

$$\phi = 4eV$$

8. **Ans. 6.32 to 6.33**

$$\text{Sol. } mvR = \left(mR^2 + \frac{MR^2}{2} \right) \omega$$

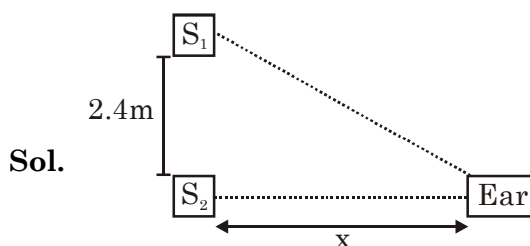
$$\omega = \frac{mvR}{mR^2 + \frac{MR^2}{2}} = \frac{V}{2R}$$

$$-mgR = 0 - \frac{1}{2} \left(mR^2 + \frac{MR^2}{2} \right) \omega^2$$

$$V = \sqrt{40}$$

$$V = 6.324 \text{ m/s}$$

9. **Ans. 7.00**



Sol.

$$f = 425 \text{ Hz}$$

$$V = 340 = f\lambda$$

$$\Rightarrow \lambda = 0.8 \text{ m}$$

$$\text{For no sound, } \Delta x = (2n+1) \frac{\lambda}{2}$$

$$\sqrt{x^2 + (2.4)^2} - x = \frac{\lambda}{2}$$

$$\Rightarrow x = 7 \text{ m}$$

10. **Ans. 8.00**

Sol. Only friction can reduce speed of particle

$$\Rightarrow W_{fr} = \text{DKE}$$

$$-\mu NS = 0 - \frac{1}{2} mu^2$$

$$\mu(mg + qE)S = 0 - \frac{1}{2} mu^2$$

$$S = \frac{mu^2}{2\mu(mg + qE)}$$

$$S = 125 = \frac{1000}{n} \Rightarrow n = 8$$

PART-2 : CHEMISTRY

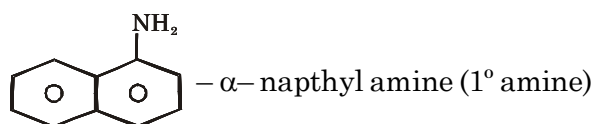
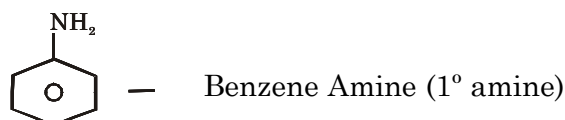
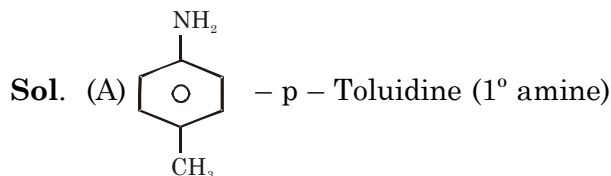
SOLUTION

SECTION-I

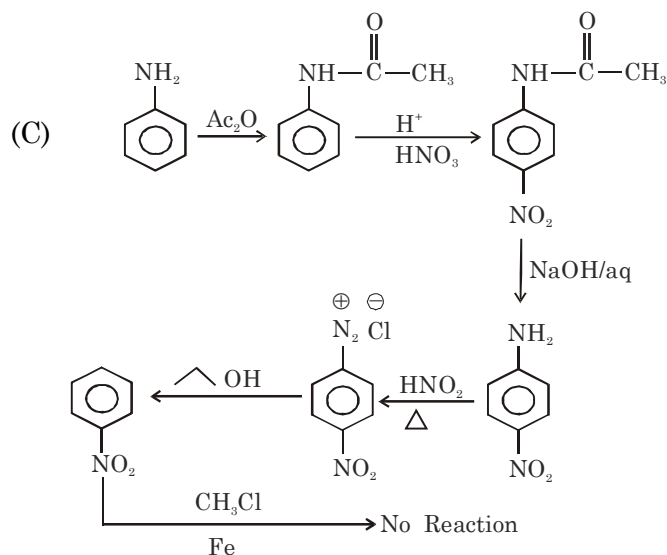
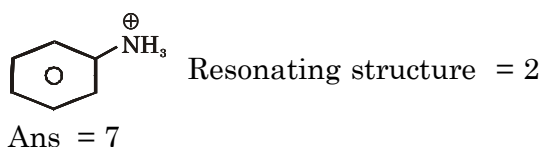
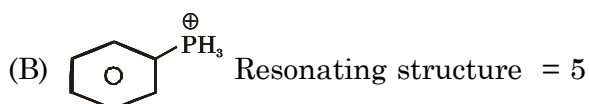
1. Ans. (C,D)

2. Ans. (B)

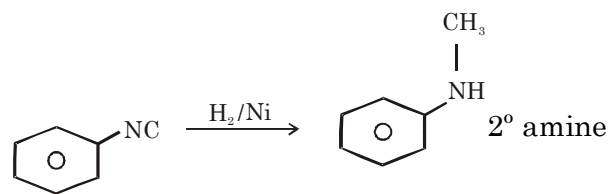
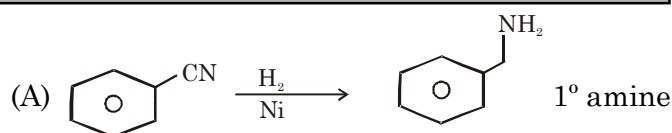
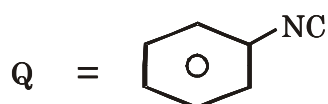
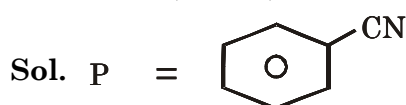
3. Ans. (A,D)



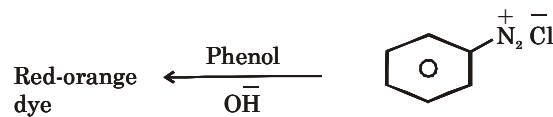
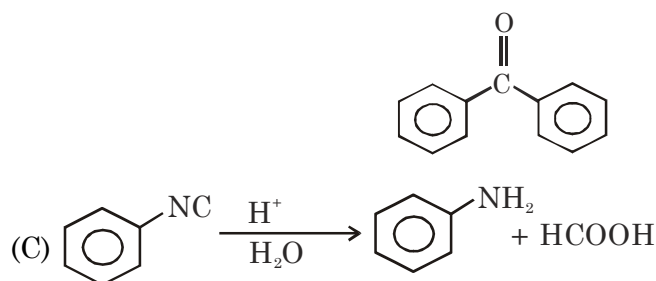
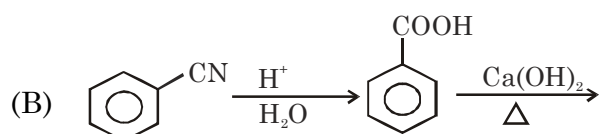
All 1° amine give Hinsberg test



4. Ans. (B,C,D)



Functional isomer not metamer



5. Ans. (A,B,C,D)

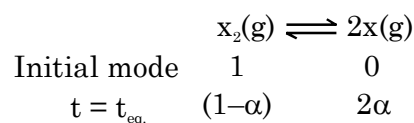
Sol. (A) Correct statement.

As on decrease in pressure reaction move in direction where no. of gaseous molecules increase

(B) Correct statement

At the start of reaction $Q_p < K_p$ so dissociation of X_2 take place spontaneously.

(C) Paragraph-1



Given $2\alpha = \beta_{\text{equilibrium}}$

So $\alpha = \frac{\beta_{\text{equilibrium}}}{2}$

Total mole at equilibrium = $(1 + \alpha) = 1 +$

$$\left(\frac{\beta_{eq.}}{2}\right)$$

$$P_{x_2} = \left[\frac{1 - \frac{\beta_{eq.}}{2}}{1 + \frac{\beta_{eq.}}{2}} P_{total} \right] = \left[\frac{2 - \beta_{eq.}}{2 + \beta_{eq.}} P_{total} \right] = \left[\frac{2 - \beta_{eq.}}{2 + \beta_{eq.}} P_{total} \right]$$

$$P_{x(g)} = \left[\frac{\beta_{eq.}}{1 + \frac{\beta_{eq.}}{2}} P_{total} \right] = \left[\frac{2\beta_{eq.}}{2 + \beta_{eq.}} \right] P_{total}$$

$$\text{So } K_p = \frac{(P_x)^2}{(P_{x_2})} = \left[\frac{\frac{2\beta_{eq.}}{2 + \beta_{eq.}} \times P_{total}}{\frac{2 - \beta_{eq.}}{2 + \beta_{eq.}} \times P_{total}} \right]^2$$

$$K_p = \frac{4\beta_{eq.}^2}{4 - \beta_{eq.}^2} \times P_{total} = \left(\frac{8\beta_{eq.}^2}{4 - \beta_{eq.}^2} \right)$$

(D) Correct statement

As $\Delta G^\circ > 0$ & $\Delta G^\circ = -RT \ln K_p$

$\Delta G^\circ > 0$, So K_p should be less than 1.

So $K_p < 1$

$$K_p = K_c(RT)^{\Delta n_g} \quad (RT > 1)$$

$$K_c = \frac{K_p}{RT}$$

$$K_c < K_p \quad \text{So } K_c < 1$$

6. Ans. (A,B,C)

7. Ans. (B)

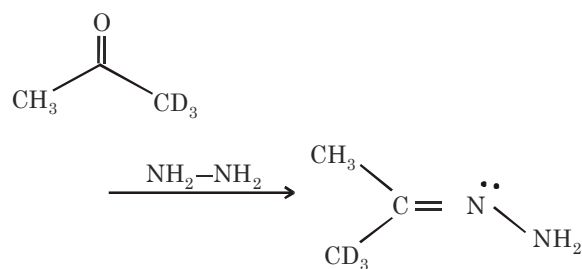
8. Ans. (A,D)

9. Ans. (D)

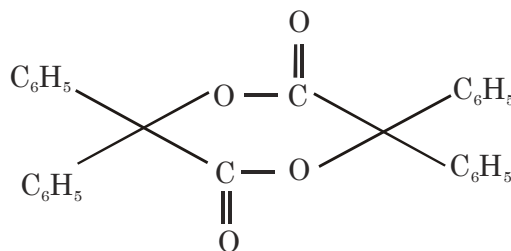
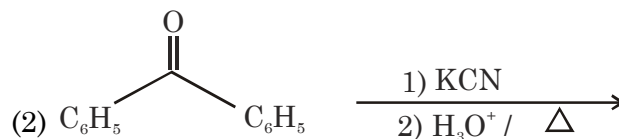
10. Ans. (C)

11. Ans. (C)

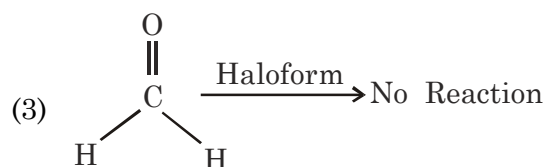
Sol. (1)



POS present



COS present



12. Ans. (C)

13. Ans. (B)

14. Ans. (A)

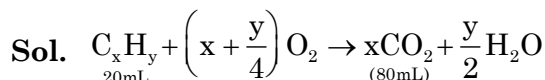
SECTION-II

1. Ans. (12.50)

Sol. Therefore, $\frac{t}{t_{1/2}} = \frac{180}{60} = 3.0$

or $N = \frac{N_0}{2^{3.0}} = \frac{1}{8} N_0 = \frac{1}{8} \times 100$ percent of
 $N_0 = 12.5\%$

2. Ans. (12.00)



$$CO_2 \Rightarrow 80 \text{ mL}$$

$$20x = 80$$

$$x = 4$$

Now

Let initial volume of $O_2 = v_1$

volume contraction is given as 60 mL

$$20 + V_1 - \left\{ 80 + \left[V_1 - \left(x + \frac{y}{4} \right) 20 \right] \right\} = 60$$

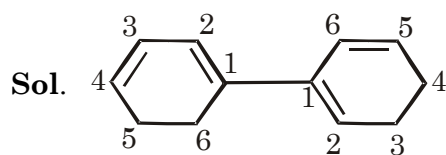
$$-60 + 80 + 5y = 60$$

$$20 + 5y = 60$$

$$5y = 40$$

$$y = 8$$

3. Ans. (10.00)



1 - (cyclo hex - 1, 5- dieny l) cyclohexa -1, 3- diene

$$\text{Sum} = 1 + 5 + 1 + 3 = 10$$

4. Ans. (33.33)

Sol. $\frac{n_{\text{He}/t}}{n_{\text{CH}_4/t}} = \frac{P_{\text{He}}}{P_{\text{CH}_4}} \sqrt{\frac{M_{\text{CH}_4}}{M_{\text{He}}}} = \frac{n_{\text{He}}}{n_{\text{CH}_4}} \sqrt{\frac{M_{\text{CH}_4}}{M_{\text{He}}}} = \frac{4}{1} \times \sqrt{\frac{16}{4}} = \frac{8}{1}$

$$\therefore \text{mass \% of CH}_4 = \frac{1 \times 16}{1 \times 16 + 8 \times 4} \times 100$$

$$= 33.33$$

5. Ans. (100.00)

Sol. $\frac{10^{-3} \times 100}{1000} \times 10^6 \Rightarrow 100 \text{ ppm.}$

6. Ans. (3.00)

7. Ans. (20.00)

8. Ans. (4.00)

9. Ans. (6.00)

Sol. Acidic group $\rightarrow$ Alkyne, Ketone, aldehyde
 $\rightarrow$ Alcohol, Amine, Acid will react with grignard reagent.

10. Ans. (7.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,C)

Sol.0412
2024
4032

$$n(A) = 6 \times 3 + 4 \times 3 = 30$$

$$1, 2, 3, 0 ; 4, 2, 3, 0$$

$$n(B) = 18 + 18 = 36$$

Using 1, 2, 3, 0

$$\dots 12, \dots 20, \dots 32$$

$$1 + 2 + 1 = 4$$

Using 4, 2, 3, 0

$$\dots 04, \dots 20, \dots 24, \dots 32, \dots 40$$

$$2 + 2 + 1 + 1 + 2 = 8$$

$$n(A \cap B) = 8$$

2. Ans. (A,B,C)

Sol. $\frac{2 \cos 5x - 2 \sin 4x + \sin 3x - \sin 7x + \cos 2x - \cos 6x}{\cos 5x - \sin 4x} = 1$

$$\Rightarrow \frac{2(\cos 5x - \sin 4x) - 2 \cos 5x \sin 2x + 2 \sin 4x \sin 2x}{\cos 5x - \sin 4x} = 1$$

$$\frac{(2 - 2 \sin 2x)[\cos 5x - \sin 4x]}{\cos 5x - \sin 4x} = 1$$

$$\Rightarrow \sin 2x = \frac{1}{2} \text{ and } \sin 4x \neq \cos 5x$$

$$2x = 2n\pi + \frac{\pi}{6} \text{ or } 2n\pi + \frac{5\pi}{6}$$

$$x = n\pi + \frac{\pi}{12} \text{ or } n\pi + \frac{5\pi}{12} \text{ where } n \in \mathbb{Z}$$

$$\Rightarrow x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}, \dots$$

3. Ans. (B,C)

Sol. $f(x) + 3x f\left(\frac{1}{x}\right) = 2(x+1) \dots (1)$

Replace x by $\frac{1}{x}$

$$f\left(\frac{1}{x}\right) + \frac{3}{x} f(x) = 2\left(\frac{1}{x} + 1\right) \dots (2)$$

Solving (1) and (2), we get

$$f(x) = \frac{x+1}{2} \quad f'(x) = \frac{1}{2}$$

$$f(x) = x^2 \Rightarrow 2x^2 - x - 1 = 0$$

$$x = 1, x = -\frac{1}{2} \text{ (Rejected)}$$

4. Ans. (A,B,C,D)

Sol. $f(x) = \sin 2\alpha (\sin(\alpha - \beta)) - \sin 2\beta (\sin(\alpha - 2))$
 $f(x) = k$

5. Ans. (A,B,C,D)

Sol. Let S'_1 be the reflection of $S_1 (3, 2)$

$$\therefore \frac{x-3}{1} = \frac{y-2}{1} = -3$$

$$S_1'(0, -1) \text{ and } S_1'S_2 : y = 2x - 1$$

solve with $x + y - 2 = 0$

$$a = 1, b = 1$$

$$PS_1 + PS_2 = \sqrt{5} + \sqrt{45}$$

$$S_1S_2 = \sqrt{26}$$

6. Ans. (A,B,C,D)

Sol. (A),(B), (C), (D) Given expression

$$E = -\frac{2}{(n+1)} \sum_{r=0}^n {}^{n+1}C_{r+1} (-\sqrt{k})^{r+1}$$

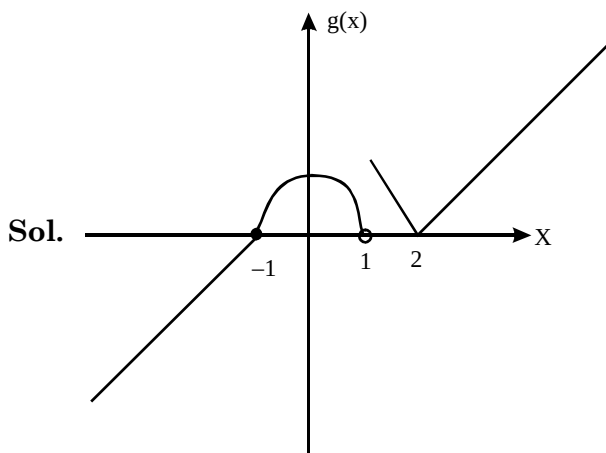
$$= \frac{2}{n+1} \left(-\left(1 - \sqrt{k}\right)^{n+1} + 1 \right).$$

Evaluate (B) and (C) by putting $\sqrt{x} = y$ and $t^2 = y$ respectively, and then integrating
In (D)

$$t_r = (-1)^r \cdot 2 \frac{{}^nC_r (\sqrt{k})^{r+1}}{r+1} = \frac{2}{n+1} {}^{n+1}C_{r+1} (-1)^r (\sqrt{k})^{r+1}$$

= general term of E.

7. Ans. (A,C,D)



$g(x)$ is discontinuous at $x=1$ and non-differentiable at $x=-1, 1, 2$

8. Ans. (A,C)

Sol. $\sin 2A = \sin 2B$

$$\Rightarrow 2A = \pi - 2B \quad (\because A \neq B)$$

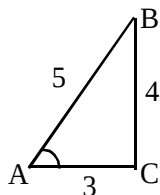
$$\Rightarrow A + B = \frac{\pi}{2}$$

$$\Rightarrow C = \frac{\pi}{2}$$

$$\tan A = \frac{4}{3}$$

$$\Rightarrow \frac{4}{1 + \tan^2 B} + 2 \sin(B - A) + \cot C (\cos B \sqrt{1 + \sin^2 A} - \cos A \sqrt{1 + \sin^2 B})$$

$$= 2$$



Solutions for 9 to 11

9. Ans. (B)

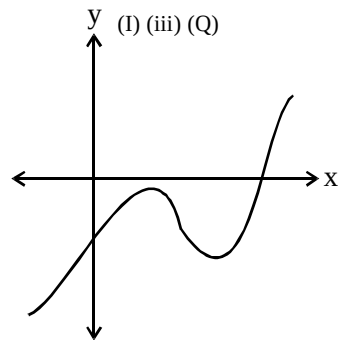
10. Ans. (D)

11. Ans. (C)

Solution :

(I). $f(x) = x^3 - 6x^2 + 9x - 12$

$$f(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$$



2 points of local Extrema

1 solution of $f(x) = 0$

(II) $f(x) = 6x - 3\sin x - 4\cos x - 2020$

$$f(x) = 6 - 3\cos x + 4\sin x > 0$$

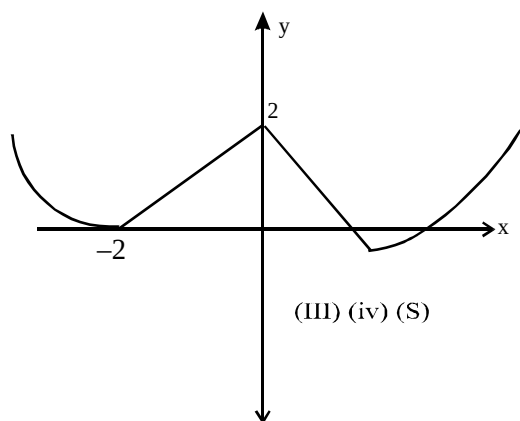
Zero points of Extrema

one solution of $f(x) = 0$

(II)(i)(Q)

(III) $f(x) = (2 - |x|, x^2 - x - 6)$

3 points of local Extrema

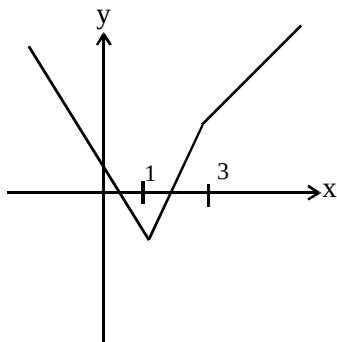


(III) (iv) (S)

3 solution of $f(x) = 0$

(IV) $f(x) = |2x - |x - 3|| - 1$

$$f(x) = \begin{cases} x + 2 & ; \quad x \geq 3 \\ 3x - 4 & ; \quad 1 \leq x < 3 \\ 2 - 3x & ; \quad x < 1 \end{cases}$$



(IV)(ii)(R)

1 Point of local Extrema

2 Solution of $f(x) = 0$

Sol. 12 to 14

12. Ans. (D)

13. Ans. (D)

14. Ans. (D)

Solution

$$(I) \quad \vec{r} = \lambda \vec{a} + \mu \vec{b} + \gamma (\vec{a} \times \vec{b})$$

$$\vec{r} \cdot \vec{a} = 0 \Rightarrow \lambda + \frac{\mu}{2} = 0$$

$$\vec{r} \cdot \vec{a} = 2 \Rightarrow \frac{\lambda}{2} + \mu = 2$$

$$(\vec{r} \times \vec{a}) \cdot \vec{b} = 1 \Rightarrow \frac{3\gamma}{4} = 1$$

$$\text{Solving } \lambda = \frac{-4}{3}, \mu = \frac{8}{3}, \gamma = \frac{4}{3}$$

(I)(iii)(R)

$$(II) \quad \vec{r} \cdot \vec{a} = 0 \Rightarrow \lambda + \frac{\mu}{2} = 0$$

$$\vec{r} \cdot \vec{b} = 1 \Rightarrow \frac{\lambda}{2} + \mu = 1$$

$$\gamma = 0; \lambda = \frac{-2}{3}, \mu = \frac{4}{3}$$

(II)(ii)(P)

$$(III) \quad \vec{r} \times \vec{a} = 3\vec{a} \times \vec{b}$$

$$(\vec{r} + 3\vec{b}) \times \vec{a} = 0$$

$$\vec{r} + 3\vec{b} = \lambda \vec{a}$$

$$\vec{r} = \lambda \vec{a} - 3\vec{b}$$

$$\vec{r} \cdot \vec{a} = 1 \Rightarrow \lambda \frac{-3}{2} = 1 \Rightarrow \lambda = \frac{5}{2}$$

$$\vec{r} = \frac{5}{2} \vec{a} - 3\vec{b}$$

$$\lambda = \frac{5}{2}, \mu = -3, \gamma = 0$$

(III)(iv)(P)

$$(IV) \quad \vec{a} \times (\vec{b} \times \vec{r}) = \vec{a} \times \vec{b}$$

$$(\vec{a} \cdot \vec{r}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{r} = \vec{a} \times \vec{b}$$

$$2\vec{b} - \frac{1}{2}\vec{r} = \vec{a} \times \vec{b}$$

$$\vec{r} = 4\vec{b} - 2(\vec{a} \times \vec{b})$$

$$\lambda = 0, \mu = 4, \gamma = 2$$

(IV)(iii)(S)

SECTION-II

1. Ans. 7.00

$$\text{Sol.} = \int \underbrace{x \sin x}_I \underbrace{e^{e^x + x^3 + x} (e^x + 3x^2 + 1)}_{II} dx$$

$$+ \int e^{e^x + x^3 + x} (\sin x + x \cos x) dx$$

$$= x \sin x e^{e^x + x^3 + x} - \int (x \cos x + \sin x) e^{e^x + x^3 + x} dx$$

$$+ \int e^{e^x + x^3 + x} (\sin x + x \cos x) dx$$

$$= x \sin x e^{e^x + x^3 + x} + c$$

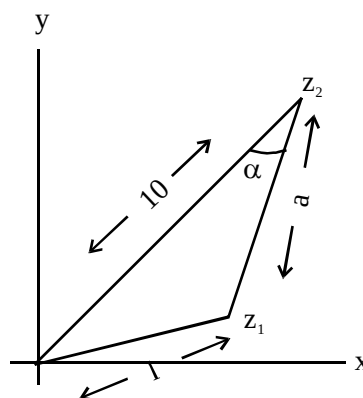
2. Ans. 1.00

$$\text{Sol.} \quad \theta = \arg \left(\frac{z_1 - z_2}{z_2} \right)$$

$$\theta = \pi + \arg \left(\frac{z_2 - z_1}{z_2} \right) = \pi + \alpha$$

$$\text{let } a = |z_1 - z_2|$$

$$\cos \alpha = \frac{a^2 + 100 - 1}{20a}$$



$$\cos \alpha = \frac{\left(a + \frac{99}{a}\right)}{20}$$

$$\tan^2 \theta = \sec^2 \theta - 1 = \sec^2 (\pi + \alpha) - 1 = \sec^2 \alpha - 1$$

$$\tan^2 \theta = \frac{400}{\left(a + \frac{99}{a}\right)^2} - 1$$

$$= \frac{400}{4 \times 99} - 1 \quad (\because A.M \geq G.M.)$$

$$= \frac{1}{99}$$

3. **Ans : 1.50**

Centre is (0, 0)

Let P(rcosθ, rsinθ) be any point on ellipse.

$$\Rightarrow r^2 = \frac{4}{9 + \cos 2\theta + \sin 2\theta}$$

$$\Rightarrow r_{\max} = \frac{2}{\sqrt{9 - \sqrt{2}}}$$

$$\therefore r^2 = \frac{36 + \sqrt{32}}{79}$$

$$a = 36, b = 32, c = 79$$

4. **Ans. 8.00**

Sol. $\int_0^{\frac{\pi}{4}} [3 \tan^2 x] dx$

$$\int_0^{\frac{\pi}{6}} 0 dx + \int_{\frac{\pi}{6}}^{\tan^{-1} \sqrt{\frac{2}{3}}} 1 dx + \int_{\tan^{-1} \sqrt{\frac{2}{3}}}^{\frac{\pi}{4}} 2 dx$$

$$0 + \left(\tan^{-1} \sqrt{\frac{2}{3}} - \frac{\pi}{6} \right) + 2 \left(\frac{\pi}{4} - \tan^{-1} \sqrt{\frac{2}{3}} \right)$$

$$= \frac{\pi}{3} - \tan^{-1} \left(\sqrt{\frac{2}{3}} \right)$$

5. **Ans. 2.00**

Sol. $\frac{x-0}{3} = \frac{y-1}{4} = \frac{z+1}{0} = \lambda$

$$\bullet A(0, 1, -1)$$



$$\bullet B(3, 5, -1)$$

General Point on line is $(3\lambda, 4\lambda + 1, -1)$

This point will satisfy plane $x + y - 2z = 5$

$$3\lambda + 4\lambda + 1 + 2 = 5$$

$$\lambda = \frac{2}{7}$$

$$P\left(\frac{6}{7}, \frac{15}{7}, -1\right)$$

6. **Ans. 2.50**

$$y \frac{dx}{dy} + 2x = y^2 \ln y$$

$$\frac{dx}{dy} + \left(\frac{2}{y}\right)x = y \ln y$$

$$I.F. = e^{\int \frac{2}{y} dy} = y^2$$

$$\Rightarrow y^2 x = \int (y \ln y) y^2 dy$$

$$xy^2 = \frac{y^4 (\ln y)}{4} - \frac{y^4}{16} + c$$

$$\text{curve passes through } \left(\frac{-1}{16}, 1\right) \Rightarrow c = 0$$

$$\Rightarrow x = \frac{y^2 \ln y}{4} - \frac{y^2}{16}$$

$$\Rightarrow 16x + y^2 = 4y^2 \ln y$$

$$a=1, b=4$$

7. **Ans. 1.50**

Sol. $\frac{(a-b)(2a-b)}{a(a-b+c)} = \frac{\left(1-\frac{b}{a}\right)\left(2-\frac{b}{a}\right)}{\left(1-\frac{b}{a}+\frac{c}{a}\right)}$

$$= \frac{(1+\alpha+\beta)(2+\alpha+\beta)}{1+\alpha+\beta+\alpha\beta}$$

$$= \frac{\alpha}{\beta+1} + \frac{\beta}{\alpha+1} + 2$$

$$\text{there } 0 \leq \alpha \leq \beta \leq 1$$

$$\Rightarrow m = 3, n = 2$$

8. Ans. 0.60

Sol. $m_{PA} = \frac{3-2}{3-4} = -1$

$\Rightarrow$ Equation of tangent at P

$y - 3 = x - 3 \Rightarrow y = x \dots(1)$

$m_{QA} = \frac{-1-2}{1-4} = 1$

$\Rightarrow$ equation of tangent at Q

$y + 1 = -(x - 1) \Rightarrow x + y = 0 \dots(2)$

Tangents at P and Q are mutually perpendicular

$\Rightarrow$ PQ is a focal chord, equation of PQ is $y = 2x - 3$

Slope of axis of the parabola $= \frac{2-0}{4-0} = \frac{1}{2}$

$\Rightarrow$ Equation of directrix $2x + y = 0$

$S(\alpha, \beta)$ lies on focal chord

$\Rightarrow \beta = 2\alpha - 3$

Image of $S(\alpha, \beta)$ with respect to $y = x$ line lies on $2x + y = 0$

$\Rightarrow 2\beta + \alpha = 0$

$\Rightarrow \alpha = \frac{6}{5}, \beta = \frac{-3}{5}$

$\Rightarrow \alpha + \beta = \frac{3}{5} = 0.60$

9. Ans. 0.50

Sol. $6x + 3y + 2z - 6 = 0$

$P(x_1, y_1, z_1)$

$A(0, y_1, z_1), B(x_1, 0, z_1), C(x_1, y_1, 0)$

$$\vec{N} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1 & -y_1 & 0 \\ 0 & y_1 & -z_1 \end{vmatrix}$$

$= (y_1 z_1) \hat{i} + (x_1 z_1) \hat{j} + (x_1 y_1) \hat{k}$

$\Rightarrow$ Equation of plane containing points A, B & C

$\frac{x}{x_1} + \frac{y}{y_1} + \frac{z}{z_1} = 2 \dots(1)$

Let $P(\alpha, \beta, \gamma)$ be the foot of perpendicular from $(0,0,0)$ to the plane (1)

$\Rightarrow \frac{\alpha}{1/x_1} = \frac{\beta}{1/y_2} = \frac{\gamma}{1/z_1} = \lambda$

$x_1 = \frac{\lambda}{\alpha}, y_1 = \frac{\lambda}{\beta}, z_1 = \frac{\lambda}{\gamma}$

(x_1, y_1, z_1) lies in $6x + 3y + 2z - 6 = 0$

$\Rightarrow \frac{6\lambda}{\alpha} + \frac{3\lambda}{\beta} + \frac{2\lambda}{\gamma} = 6 \dots(2)$

(α, β, γ) lies in (1)

$\Rightarrow \frac{\alpha^2}{\lambda} + \frac{\beta^2}{\lambda} + \frac{\gamma^2}{\lambda} = 2 \dots(3)$

from (2) & (3)

$(\alpha^2 + \beta^2 + \gamma^2) \left(\frac{6}{\alpha} + \frac{3}{\beta} + \frac{2}{\gamma} \right) = 12$

$\Rightarrow$ locus is $(x^2 + y^2 + z^2) \left(\frac{1}{x} + \frac{1}{2y} + \frac{1}{3z} \right) = 2$

10. Ans. 0.15

Sol. $L = \lim_{n \rightarrow \infty} \sum_{K=1}^n \frac{\frac{K^4}{n^5} + \frac{K^9}{n^{10}}}{\sqrt{1 + \frac{K^4}{n^5} + \frac{K^9}{n^{10}} + 1}}$

Use Sandwich

$\frac{\int_0^1 (x^4 + x^9) dx}{2} \leq L \leq \frac{\int_0^1 (x^4 + x^9) dx}{2}$

$\Rightarrow L = \frac{\frac{1}{5} + \frac{1}{10}}{2} = 0.15$